STATISTICS IS THE GRAMMAR OF SCIENCE

PROBABILITY AND STATISTICS

LECTURE – 18

PROBABILITY DISTRIBUTIONS

BETA DISTRIBUTION

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BETA DISTRIBUTION

Beta distribution was first introduced by **Reverend Thomas Bayes.** The beta distribution is frequently used to model any data that occur over an interval because we know that because, as we know that, any interval can be transformed to a (0, 1) interval. Actually we can say that *Beta distribution is an extension of uniform distribution*.

THE BETA DISTRIBUTION A continuous random variable X is said to have a beta distribution if it has the following probability density function

$$f(x)=\begin{cases}\dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1} & for\ 0<x<1\\ 0 & otherwise\end{cases}$$

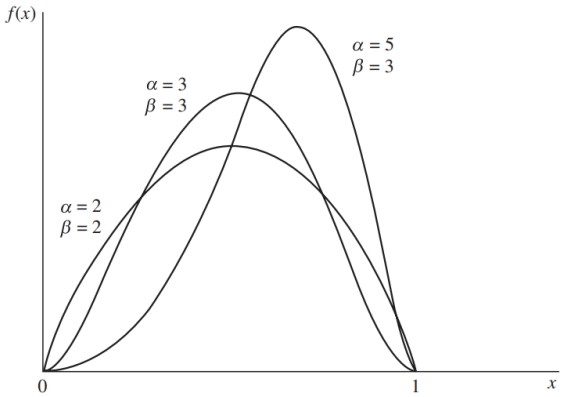
$$where\ the\ parameter\ \alpha>0\ \ and\ \ \beta>0.$$

Note that p.d.f of beta distribution can also be defined in terms of beta function

$$f(x)=\begin{cases}\dfrac{1}{B(\alpha,\beta)}x^{\alpha-1}(1-x)^{\beta-1} & for\ 0<x<1\\ 0 & otherwise\end{cases}$$

Where $$B(\alpha,\beta)=\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx=\frac{\Gamma(\alpha).\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

GRAPHS OF SOME BETA DISTRIBUTIONS



PROPERTIES OF BETA DISTRIBUTION

1- The Mean of Beta Distribution is $E(X)=\alpha/\alpha+\beta$

2- The variance of Beta Distribution is $V(X)=\alpha\beta/(\alpha+\beta)^2(\alpha+\beta+1)$

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EXAMPLES OF BETA DISTRIBUTION

EXAMPLE-1 The proportion of the stocks that will increase in value tomorrow is a beta random variable with parameters α and β. Today, these parameters were determined to be α = 5 and β = 4, by a financial analyst, after she assessed the current political, social, economical, and financial factors. What is the probability that tomorrow the values of at least 70% of the stocks will move up?

SOLUTION Let X be the proportion of the stocks that will move up in value tomorrow

$Here\ \alpha = 5\ \ and\ \ \beta = 4.\ we\ have\ to\ \ find\ P(X \geq 0.70) = ?$

$$Now\ P(X \geq 0.70) = \int_{0.70}^{1} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}dx$$

$$= \int_{0.70}^{1} \frac{\Gamma(5+4)}{\Gamma(5)\Gamma(4)} x^{5-1}(1-x)^{4-1}dx$$

$$= \int_{0.70}^{1} \frac{\Gamma(9)}{\Gamma(5)\Gamma(4)} x^{4}(1-x)^{3}dx$$

$$= \int_{0.70}^{1} \frac{8!}{4!\times 3!} x^{4}(1-x)^{3}dx$$

$$= 280\int_{0.70}^{1} x^{4}(1-x)^{3}dx = 0.194$$

EXAMPLE-2 When a queen bee leaves a bee colony to start a new hive, a certain proportion of the worker bees take flight and follow her. An entomologist models the proportion *X* of the worker bees that leave with the queen using a beta distribution with parameters $\alpha = 2\ \ and\ \ \beta = 3$. Find the probability that more than half of the bee colony leaves with the queen.

SOLUTION Let X denote the proportion *X* of the worker bees that leave with the queen

$Here\ \alpha = 2\ \ and\ \ \beta = 3.\ we\ have\ to\ \ find\ \ P(X > 0.5) = ?$

$$Now\ P(X > 0.5) = \int_{0.5}^{1} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}dx$$

$$= \int_{0.5}^{1} \frac{\Gamma(2+3)}{\Gamma(2)\Gamma(3)} x^{2-1}(1-x)^{3-1}dx$$

$$= \int_{0.5}^{1} \frac{\Gamma(5)}{\Gamma(2)\Gamma(3)} x^{1}(1-x)^{2}dx$$

$$= \int_{0.5}^{1} \frac{4!}{1!\times 2!} x^{1}(1-x)^{2}dx$$

$$= 12\int_{0.5}^{1} x(1-x)^{2}dx = 0.3125$$

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EXAMPLE-3 A gasoline wholesale distributor uses bulk storage tanks to hold a fixed supply. The tanks are filled every Monday. Of interest to the wholesaler is the proportion of the supply sold during the week. Over many weeks, this proportion has been observed to match fairly well a beta distribution with $\alpha = 4$ and $\beta = 2$.

(a) Find the expected value of this proportion.

(b) Find the variance value of this proportion.

(c) Is it highly likely that the wholesaler will sell at least 90% of the stock in a given week?

SOLUTION Let X denote the proportion of the total supply sold in a given week

$Here\ \alpha = 4 \quad and \quad \beta = 2.$

$$(a)\ E(X) = \frac{\alpha}{\alpha+\beta} = \frac{4}{4+2} = \frac{4}{6} = \frac{2}{3}$$

$$(b)\ V(X) = \frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$$

$$= \frac{(4)(2)}{(4+2+1)(4+2)^2}$$

$$= \frac{(8)}{(7)(36)} = 0/0317$$

$(b)\ Here\ \alpha = 4 \quad and \quad \beta = 2.\ we\ have\ to\ \ find\ P(X > 0.9) = ?$

$$P(X > 0.9) = \int_{0.9}^{1} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1} dx$$

$$= \int_{0.9}^{1} \frac{\Gamma(4+2)}{\Gamma(4)\Gamma(2)} x^{4-1}(1-x)^{2-1} dx$$

$$= \int_{0.9}^{1} \frac{\Gamma(6)}{\Gamma(4)\Gamma(2)} x^{3}(1-x)^{1} dx$$

$$= \int_{0.9}^{1} \frac{5!}{3!\times 1!} x^{3}(1-x)^{1} dx$$

$$= 20\int_{0.9}^{1} x^{3}(1-x)^{1} dx = 0.08$$

Therefore, it is not very likely that 90% of the stock will be sold in a given week

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IMPORTANT CONCEPTS AND FUNCTIONS IN PROBABILITY

GAMMA FUNCTION

$$\Gamma(n)=\int_0^{\infty} x^{n-1}e^{-x}dx \qquad \text{also} \qquad \int_0^{\infty} x^{n-1}e^{-ax}dx=\frac{\Gamma(n)}{a^n}$$

PROPERTIES OF GAMMA FUNCTION

1. $\Gamma(n)=(n-1).\Gamma(n-1)$
2. $\Gamma(n)=(n-1)!$ *If* 'n' *is a positive* int*eger.*
3. $\Gamma(1)=1$
4. $\Gamma(\frac{1}{2})=\sqrt{\pi}$
5. $\Gamma(-\frac{1}{2})=\frac{1}{\sqrt{\pi}}$

EXAMPLES OF GAMMA FUNCTION

$$\int_0^{\infty} x\ e^{-x}\,dx=\Gamma(2)=1!=1$$

$$\int_0^{\infty} x^2\ e^{-x}\,dx=\Gamma(3)=2!=2$$

$$\int_0^{\infty} x^3\ e^{-2x}\,dx=\frac{\Gamma(4)}{2^4}=\frac{3!}{16}=\frac{6}{16}=\frac{3}{8}$$

$$\int_0^{\infty} x^5\ e^{-3x}\,dx=\frac{\Gamma(6)}{3^6}=\frac{5!}{729}=\frac{120}{729}$$

BETA FUNCTION

$$B(\alpha,\beta)=\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx=\frac{\Gamma(\alpha).\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

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PROPERTIES OF BETA DISTRIBUTION

PROPERT-1 AREA UNDER THE BETA CURVE

We have to show that $\int_0^1 f(x)dx = 1$

$$L.H.S = \int_0^1 f(x)dx$$

$$= \int_0^1 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \times \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$$= 1 = R.H.S$$

PROPERT-2 MEAN OF BETA DISTRIBUTION

$$Mean = E(X) = \int_0^1 x\, f(x)dx$$

$$= \int_0^1 x \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{1+\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{\alpha}(1-x)^{\beta-1}\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \times \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \times \frac{\alpha\,\Gamma(\alpha)\Gamma(\beta)}{(\alpha+\beta)\Gamma(\alpha+\beta)}$$

$$= \frac{\alpha}{\alpha+\beta}$$

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PROPERT-3 VARIANCE OF EXPONENTIAL DISTRIBUTION

$$V(X)=E\left(X^2\right)-\left(E(X)\right)^2 \text{ --------(1)}$$

we have $E(X)=\dfrac{\alpha}{\alpha+\beta}$

Now $E\left(X^2\right)=\int_0^\infty x^2\,\dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}\,dx$

$$=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\int_0^1 x^{2+\alpha-1}(1-x)^{\beta-1}\,dx$$

$$=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\int_0^1 x^{\alpha+1}(1-x)^{\beta-1}\,dx$$

$$=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\times\frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)}$$

$$=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\times\frac{(\alpha+1)\Gamma(\alpha+1)\Gamma(\beta)}{(\alpha+\beta+1)\Gamma(\alpha+\beta+1)}$$

$$=\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}\times\frac{(\alpha+1)\alpha\,\Gamma(\alpha)\Gamma(\beta)}{(\alpha+\beta+1)(\alpha+\beta)\Gamma(\alpha+\beta)}$$

$$=\frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)}$$

Put the values of $E(X)$ *and* $E\left(X^2\right)$ *in equation* (1), *we get*

$$V(X)=\frac{\alpha(\alpha+1)}{(\alpha+\beta+1)(\alpha+\beta)}-\left(\frac{\alpha}{\alpha+\beta}\right)^2$$

$$=\frac{\alpha(\alpha+1)(\alpha+\beta)-\alpha^2(\alpha+\beta+1)}{(\alpha+\beta+1)(\alpha+\beta)^2}$$

$$=\frac{\alpha^3+\alpha^2\beta+\alpha^2+\alpha\beta-\alpha^3-\alpha^2\beta-\alpha^2}{(\alpha+\beta+1)(\alpha+\beta)^2}$$

$$=\frac{\alpha\beta}{(\alpha+\beta+1)(\alpha+\beta)^2}$$

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